

Recipe (locked per branch)

- structure
- atom typing
- force field
- gas model
- mixing rule
- electrostatics
- backend
- sampling protocol
- reference observable

Widom insertion

- $\langle e^{-\beta U} \rangle \rightarrow K_H, Q_{st}$

Verdict (exactly one)

- strict pass
- Tier-B physical pass
- force-field disagreement
- reference mismatch
- structural blocker
- ensemble / method mismatch

Schematic - no data values